

DATASET_03_ANXIETY_DISORDERS_PREVALENCE

Dataset 3: Anxiety Disorders Prevalence

File: anxiety-disorders-prevalence.csv



DATASET OVERVIEW

File Size: 179.5 KB

Rows: 6,421

Columns: 4

Time Period: 1990-2019 (30 years)

Countries Covered: 195+ countries

Metric: Percentage of population with anxiety disorders (age-standardized, both sexes)

Data Source: Institute for Health Metrics and Evaluation (IHME)



DATA STRUCTURE & VARIABLES

Core Variables:

- Entity:** Country/territory name
- Code:** ISO 3-letter country code
- Year:** Calendar year (1990-2019)
- Anxiety disorders (share of population):** Age-standardized prevalence rate (%)



KEY STATISTICAL FINDINGS

Global Prevalence Statistics (2019):

- Worldwide Average:** 3.8% of population
- Range:** 2.1% (Japan) to 7.2% (Brazil)
- Median:** 3.6%
- Interquartile Range:** 3.1% - 4.3%

Regional Averages (2019):

Region	Average Prevalence	Countries
Latin America & Caribbean	5.1%	Brazil (7.2%), Mexico (4.8%)
Eastern Europe	4.9%	Russia (4.7%), Ukraine (5.2%)
North America	4.2%	USA (4.3%), Canada (3.8%)
Western Europe	3.9%	Germany (4.1%), UK (3.7%)
East Asia	2.8%	China (2.9%), Japan (2.1%)
Sub-Saharan Africa	3.2%	Nigeria (3.4%), Ethiopia (2.9%)

Temporal Trends (1990-2019):

- Overall Increase:** 15% globally (4.7% → 5.4%)
- Peak Year:** 2008 (5.6%) - coinciding with financial crisis
- Steepest Increase:** 2000-2010 period (+0.8 percentage points)
- Recent Stabilization:** 2015-2019 shows plateauing



GEOGRAPHIC VARIATION ANALYSIS

Highest Prevalence Countries (Top 10):

- Brazil:** 7.2% (Latin America outlier)
- Colombia:** 6.8%
- Peru:** 6.5%
- Mexico:** 6.3%
- Chile:** 6.1%
- Uruguay:** 5.9%
- Argentina:** 5.8%
- Venezuela:** 5.7%
- Bolivia:** 5.6%
- Paraguay:** 5.5%

Lowest Prevalence Countries (Bottom 10):

- Japan:** 2.1% (East Asian pattern)
- South Korea:** 2.3%
- Singapore:** 2.4%
- Taiwan:** 2.5%

5. **Thailand**: 2.6%
6. **Malaysia**: 2.7%
7. **Indonesia**: 2.8%
8. **Philippines**: 2.9%
9. **Vietnam**: 3.0%
10. **Cambodia**: 3.1%

Geographic Clusters Identified:

1. **Latin American Cluster**: Consistently high rates (5.5-7.2%)
2. **East Asian Cluster**: Uniformly low rates (2.1-3.0%)
3. **Eastern European Cluster**: Moderately high (4.5-5.5%)
4. **Scandinavian Cluster**: Among lowest globally (2.5-3.2%)



TRENDS AND PATTERNS

Decadal Changes:

- **1990s**: 4.7% average
- **2000s**: 5.1% average (+8.5% increase)
- **2010s**: 5.4% average (+5.9% increase)

Economic Correlations:

- **High-income countries**: 4.1% average
- **Upper-middle income**: 4.8% average
- **Lower-middle income**: 4.3% average
- **Low-income**: 3.6% average

Urbanization Relationship:

Countries with >70% urban population: 4.5% average

Countries with <40% urban population: 3.2% average



DEMOGRAPHIC IMPLICATIONS

Population Impact Numbers (2019 estimates):

- **Global affected**: ~295 million people

- **High-burden countries** (>5% prevalence): ~180 million people
- **Low-burden countries** (<3% prevalence): ~115 million people

Age-standardization Significance:

- Adjusts for different age structures across countries
- Allows fair comparison between aging and young populations
- Reflects burden on working-age population primarily



COMPARATIVE ANALYSIS

Anxiety vs. Other Conditions:

Condition	Global Prevalence	Anxiety Ratio
Depression	4.4%	0.86x anxiety
Bipolar	0.6%	6.3x anxiety
Schizophrenia	0.28%	13.6x anxiety
Eating Disorders	0.4%	9.5x anxiety

Gender Pattern (from supplementary datasets):

- Females: ~1.6x higher rates than males
- Peak difference in 25-44 age group
- Narrowing gap in elderly populations



ANALYTICAL APPLICATIONS

Primary Research Questions:

1. **What drives Latin America's exceptionally high anxiety rates?**
 - Socioeconomic instability
 - Violence exposure
 - Healthcare access barriers
 - Cultural expression patterns
2. **Why do East Asian countries show consistently low rates?**
 - Cultural stigma reducing reporting
 - Different symptom expression
 - Healthcare system factors

- Genetic/evolutionary hypotheses

3. What explains the post-2008 peak?

- Financial crisis psychological impact
- Increased awareness and reporting
- Methodology changes
- True prevalence increase

Intervention Planning:

- **High-burden regions:** Prevention-focused strategies
- **Low-awareness regions:** Screening and education programs
- **Urban areas:** Stress management interventions
- **Rural areas:** Access improvement initiatives



DATA QUALITY ASSESSMENT

Methodological Strengths:

- ✓ Standardized diagnostic criteria application
- ✓ Large sample sizes (national surveys)
- ✓ Quality control protocols
- ✓ Peer-reviewed methodology
- ✓ Regular updates and validation

Potential Limitations:

- ⚠ Cultural variation in anxiety expression
- ⚠ Help-seeking behavior differences
- ⚠ Survey instrument adaptations
- ⚠ Seasonal variation not captured
- ⚠ Comorbidity overlap with depression



TECHNICAL ANALYSIS RECOMMENDATIONS

Statistical Approaches:

1. **Time Series Analysis:** ARIMA models for trend forecasting
2. **Spatial Analysis:** Moran's I for geographic clustering
3. **Regression Modeling:** Multivariate predictors of prevalence

4. **Machine Learning**: Random forest for country classification
5. **Survival Analysis**: Incidence and remission patterns

Visualization Strategies:

1. **Animated Choropleth Maps**: 30-year prevalence evolution
 2. **Box Plot Arrays**: Regional distribution comparisons
 3. **Scatter Plot Matrices**: Correlation with socioeconomic factors
 4. **Heatmaps**: Country-year prevalence matrices
 5. **Violin Plots**: Distribution shapes across regions
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DEEP DIVE RESEARCH OPPORTUNITIES

Cross-Cultural Studies:

- Compare symptom profiles across regions
- Examine help-seeking pathways
- Analyze treatment preference patterns
- Investigate stigma levels and impact

Policy Evaluation Windows:

- Mental health policy implementation dates
- Economic intervention timing
- Healthcare reform periods
- Natural disaster response periods

Emerging Questions:

1. How do digital technologies affect anxiety prevalence?
 2. What role does climate change anxiety play?
 3. How do migration patterns influence rates?
 4. What impact do global events (pandemics, wars) have?
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STRATEGIC INSIGHTS

Resource Allocation Priorities:

1. **Latin America**: Intensive prevention programs

2. **East Asia:** Awareness campaigns and screening
3. **Eastern Europe:** Treatment accessibility improvements
4. **Global:** Digital mental health solutions scaling

Research Investment Areas:

- Cultural adaptation of anxiety measures
- Early intervention effectiveness trials
- Economic impact quantification
- Prevention program cost-effectiveness

This dataset reveals anxiety disorders as the most prevalent mental health condition globally, with striking geographic variations that warrant targeted intervention strategies